

Experiment 3

AIM: Develop Expense Tracking Module with Exception Handling

Source: Program.cs

Code:

```
using System;
using System.Collections.Generic;

class Expense
{
    public string Category { get; set; }
    public double Amount { get; set; }

    public Expense(string category, double amount)
    {
        Category = category;
        Amount = amount;
    }

    public void Display()
    {
        Console.WriteLine($"Category: {Category}, Amount: ₹{Amount}");
    }
}

class ExpenseTracker
{
    static List<Expense> expenses = new List<Expense>();

    static void Main(string[] args)
    {
        while (true)
        {
            Console.WriteLine("\n===== Expense Tracking System =====");
            Console.WriteLine("1. Add Expense");
            Console.WriteLine("2. View Expenses");
            Console.WriteLine("3. Calculate Total Expense");
            Console.WriteLine("4. Exit");
            Console.Write("Enter your choice: ");

            try
            {
                int choice = Convert.ToInt32(Console.ReadLine());

                switch (choice)
                {
                    case 1:
                        AddExpense();
                        break;

                    case 2:
                        ViewExpenses();
                        break;
                }
            }
        }
    }

    static void AddExpense()
    {
        Console.WriteLine("Enter Category: ");
        string category = Console.ReadLine();

        Console.WriteLine("Enter Amount: ");
        double amount = Convert.ToDouble(Console.ReadLine());

        Expense expense = new Expense(category, amount);
        expenses.Add(expense);
        Console.WriteLine("Expense added successfully.");
    }

    static void ViewExpenses()
    {
        if (expenses.Count == 0)
        {
            Console.WriteLine("No expenses found.");
        }
        else
        {
            Console.WriteLine("Viewing Expenses:");
            foreach (Expense expense in expenses)
            {
                expense.Display();
            }
        }
    }

    static void CalculateTotalExpense()
    {
        double total = 0;
        foreach (Expense expense in expenses)
        {
            total += expense.Amount;
        }
        Console.WriteLine($"Total Expense: ₹{total}");
    }
}
```

```
        case 3:
            CalculateTotal();
            break;

        case 4:
            Console.WriteLine("Exiting...");
            return;

        default:
            Console.WriteLine("Invalid choice! Please select 1-4.");
            break;
    }
}
catch (FormatException)
{
    Console.WriteLine("Error: Please enter a valid number.");
}
catch (Exception ex)
{
    Console.WriteLine("Unexpected Error: " + ex.Message);
}
}

static void AddExpense()
{
    try
    {
        Console.Write("Enter Expense Category: ");
        string category = Console.ReadLine();

        Console.Write("Enter Expense Amount: ");
        double amount = Convert.ToDouble(Console.ReadLine());

        if (amount < 0)
            throw new ArgumentException("Expense amount cannot be negative.");

        expenses.Add(new Expense(category, amount));
        Console.WriteLine("Expense added successfully.");
    }
    catch (FormatException)
    {
        Console.WriteLine("Error: Amount must be a numeric value.");
    }
    catch (ArgumentException ex)
    {
        Console.WriteLine("Error: " + ex.Message);
    }
}

static void ViewExpenses()
{
    if (expenses.Count == 0)
    {
        Console.WriteLine("No expenses found.");
        return;
    }
}
```

```
        Console.WriteLine("\n----- Expense List -----");
        foreach (Expense expense in expenses)
        {
            expense.Display();
        }
    }

    static void CalculateTotal()
    {
        double total = 0;

        foreach (Expense expense in expenses)
        {
            total += expense.Amount;
        }

        Console.WriteLine($"Total Expense: ₹{total}");
    }
}
```

Output: Screenshot:

===== Expense Tracking System =====

1. Add Expense
2. View Expenses
3. Calculate Total Expense
4. Exit
Enter your choice: 1

Enter Expense Category: Food
Enter Expense Amount: 500
Expense added successfully.

===== Expense Tracking System =====

1. Add Expense
2. View Expenses
3. Calculate Total Expense
4. Exit
Enter your choice: 1

Enter Expense Category: Transport
Enter Expense Amount: 250
Expense added successfully.

===== Expense Tracking System =====

1. Add Expense
2. View Expenses
3. Calculate Total Expense
4. Exit
Enter your choice: 2

----- Expense List -----

Category: Food, Amount: ₹500
Category: Transport, Amount: ₹250

===== Expense Tracking System =====

1. Add Expense
2. View Expenses
3. Calculate Total Expense
4. Exit

Enter your choice: 3

Total Expense: ₹750

===== Expense Tracking System =====

1. Add Expense
2. View Expenses
3. Calculate Total Expense
4. Exit

Enter your choice: 4

Exiting...

```
===== Expense Tracking System =====
```

- ```
1. Add Expense
2. View Expenses
3. Calculate Total Expense
4. Exit
```

```
Enter your choice: 5
```

```
Invalid choice! Please select 1-4.
```

```
===== Expense Tracking System =====
```

- ```
1. Add Expense
2. View Expenses
3. Calculate Total Expense
4. Exit
```

```
Enter your choice: 1
```

```
Enter Expense Category: Shopping
```

```
Enter Expense Amount: abc
```

```
Error: Amount must be a numeric value.
```

```
===== Expense Tracking System =====
```

1. Add Expense
2. View Expenses
3. Calculate Total Expense
4. Exit

```
Enter your choice: 1
```

```
Enter Expense Category: Fuel
```

```
Enter Expense Amount: -500
```

```
Error: Expense amount cannot be negative.
```

```
===== Expense Tracking System =====
```

1. Add Expense
2. View Expenses
3. Calculate Total Expense
4. Exit

```
Enter your choice: 2
```

```
No expenses found.
```

```
===== Expense Tracking System =====
```

1. Add Expense
2. View Expenses
3. Calculate Total Expense
4. Exit

```
Enter your choice: 3
```

```
Total Expense: ₹0
```

GitHub URL:

<https://github.com/Preet1926/Practical-3>